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Ship-Integrated Marine Cloud Brightening

Restoring Lost Aerosol Radiative Forcing via Exhaust-Plume–Entrained Salt Aerosol from Existing Scrubber Infrastructure

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This document describes a commercially viable, regulation-ready method for restoring the cloud-albedo cooling effect lost following IMO 2020 sulfur regulations, using benign salt aerosol delivered through existing ship exhaust infrastructure.

1. Executive Summary

International shipping has been inadvertently brightening marine clouds for over a century. When ships burn fuel, aerosol particles in their exhaust rise into the marine boundary layer and act as seeds, called cloud condensation nuclei (CCN), around which water vapor condenses to form cloud droplets. More seeds mean more numerous, smaller droplets, which scatter sunlight more effectively and make the cloud brighter. The result is visible from space: bright linear cloud features, called ship tracks, trail behind vessels across every major ocean corridor, documented in satellite imagery since the 1960s.[19]

In 2020, the IMO sulfur cap reduced the maximum sulfur content in marine fuels from 3.5% to 0.5%, cutting shipping SO^2 emissions by approximately 80%.[1] This was an unambiguous public health success. But it also removed the sulfate aerosol particles that had been seeding those brighter clouds. Multiple independent studies estimate the resulting positive radiative forcing—the warming effect of losing that cloud reflectivity—at +0.1 to +0.2 W/m^2 globally,[2][3][4] equivalent to approximately 2–3 years of additional greenhouse warming.[5]

This whitepaper proposes a technically simple, commercially self-sustaining method to restore the lost cloud-albedo cooling by injecting fine salt aerosol into ship exhaust plumes using low-power nozzles integrated into existing scrubber stacks. The approach introduces no new science. It combines three well-understood phenomena: the Twomey effect on cloud albedo,[6] the turbulent mixing and thermal lofting properties of ship exhaust plumes, and the proven capacity of salt particles to act as CCN in marine stratocumulus.[7][8]

The system is designed to be funded through the IMO Net-Zero Framework's pricing mechanism, creating a circular revenue model: the IMO Net-Zero Fund compensates participating shippers for verified radiative forcing, offsetting their GHG compliance costs while delivering measurable planetary cooling.

2. Problem Statement

2.1 The IMO 2020 Cooling Gap

Prior to 2020, global shipping emitted roughly 10 Tg SO^2 per year—approximately 13% of anthropogenic SO^2 emissions.[1] This sulfate aerosol brightened marine clouds across major shipping corridors, producing a net negative radiative forcing (cooling). IMO 2020 reduced these emissions to approximately 2 Tg SO^2/yr, yielding an estimated effective radiative forcing of +0.12 to +0.2 W/m^2 over the global ocean.[2][3][4]

Ship track frequency declined dramatically: machine-learning cataloging of over one million ship tracks confirmed a large reduction in cloud perturbations following IMO 2020.[9] Diamond (2023) detected reduced cloud brightening within the southeastern Atlantic shipping corridor using counterfactual statistical methods, estimating the local instantaneous forcing at O(1 W/m^2) within the corridor itself.[10]

2.2 The Opportunity

We lost cooling we did not intend to create. Nephele proposes to put it back—deliberately, cleanly, and verifiably—using a particle (NaCl) that has approximately twice the hygroscopicity of ammonium sulfate ($\kappa \approx 1.28$ vs. $\kappa \approx 0.61$), meaning it activates as a CCN more readily at a given particle size,[30] and is entirely benign to marine ecosystems.

3. Technical Approach

3.1 Principle of Operation

The system injects a fine brine spray into the ship's exhaust plume immediately downstream of the scrubber outlet. The beauty in this solution is that the exhaust plume does all the hard work:

- Turbulent mixing disperses the spray across the plume cross-section within seconds of injection, breaking up droplets and ensuring homogeneous distribution of salt nuclei.
- Droplet evaporation during plume dilution. Post-scrubber exhaust is typically saturated with water vapor from the scrubbing process. Droplets injected into this saturated plume do not dry immediately at the stack exit. However, as the plume rises and entrains ambient marine air (typically 70–80% RH), it dilutes and the relative humidity drops below saturation. Evaporation then proceeds progressively during the minutes-long transit from stack to cloud base. The target dry crystal size range is 30–1000 nm diameter—identified by Connolly et al. (2014) and Latham et al. (2012) as the CCN-active range. Achieving this distribution reliably is a key engineering objective; the primary control variable is the initial droplet size set by nozzle selection.
- Thermal buoyancy of the warm exhaust plume provides a free convective lift, carrying salt aerosol up to and into the marine boundary layer (MBL), typically 500–1500 m above sea level, without any additional energy input.
- Once entrained in the MBL, salt particles activate as CCN and increase cloud droplet number concentration (Nd), increasing cloud albedo via the well-established Twomey effect.

This piggybacks entirely on existing scrubber architecture. The scrubber stack, exhaust gas flow, and thermal profile are already in place on approximately 5,000 vessels worldwide. The only addition is a compact spray injection module. Critically, the concentration of salt aerosol delivered to the boundary layer is directly tuneable by adjusting the injection rate at the nozzle array—more brine flow produces more CCN, less produces less. This gives operators precise, real-time control over the intensity of cloud brightening, enabling adjustment based on meteorological conditions, corridor location, or regulatory requirements.

3.2 Nozzle and Injection Hardware

The injection module consists of an array of low-power pressure-atomizing nozzles mounted in a ring configuration at the scrubber stack exit. Key specifications:

Parameter	Specification
Nozzle body material	Corrosion-resistant nickel alloy (e.g. Inconel or Nitronic series)
Target droplet size (exit)	10–50 µm (target: 30–200 nm dry NaCl)
Feed solution	Seawater (~35 g/L NaCl; pre-filtered by scrubber system)
Flow rate per nozzle	0.5–2.0 L/min (tuneable)
Power consumption	<2 kW per nozzle array

The nozzle body requires a material that is practically corrosion-free in seawater, resistant to pitting, crevice corrosion, and chloride-ion stress-corrosion cracking. Nickel-chromium-molybdenum alloys such as the Inconel series (e.g. alloy 625) and nitrogen-strengthened austenitic stainless steels such as Nitronic 60 are well-established in marine exhaust, seawater, and desalination applications and meet these requirements.[11] Final alloy selection will be determined during prototype development based on operating temperature, flow velocity, and cost optimisation.[12]

3.3 Installation and Maintenance Access

Scrubber systems are subject to regular inspection under MARPOL Annex VI, with surveys at annual, intermediate, and renewal (five-year) intervals. Scrubber nozzles and internal components are routinely inspected and cleaned by ship's crew as part of planned maintenance, with enclosed-space entry procedures followed for internal access. Shore technicians calibrate sensors on a quarterly cycle. This means the scrubber stack exit—where Nephele's injection module would be mounted—is an area that is already regularly accessed for maintenance. Installation of the injection module can be performed during a scheduled port call or routine maintenance window without requiring drydocking.

3.4 Exhaust Chemistry and HCl Corrosion

A legitimate question is whether the brine spray could react with residual exhaust gas constituents. In the marine atmosphere, sea salt aerosol is known to undergo chloride depletion through reaction with sulfuric acid (H_2SO_4) and nitric acid (HNO_3), releasing hydrochloric acid gas (HCl). This is a well-documented atmospheric process that occurs naturally over hours to days as sea salt ages in the ambient atmosphere. In the concentrated exhaust plume, where residual SO_2 and NO^8 may be present (particularly downstream of open-loop scrubbers that do not fully remove these species), this reaction could occur more rapidly, potentially depleting some of the injected chloride and releasing small quantities of HCl. We note three mitigating factors: (1) post-scrubber SO_2 concentrations are already substantially reduced—compliance with MARPOL Annex VI requires exhaust SO_2/CO_2 ratios equivalent to 0.5% sulfur fuel or lower; (2) any HCl produced would be rapidly diluted in the open atmosphere and is orders of magnitude below levels of health concern at plume-dilution distances; and (3) the primary effect of chloride depletion is to convert some NaCl particles to sodium sulfate or sodium nitrate particles, both of which are also hygroscopic and can still function as CCN, albeit with lower κ values. Nonetheless, the extent of in-plume chloride depletion and its effect on CCN activation efficiency is an open question that must be quantified during prototype testing.

Regarding HCl corrosion risk to the ship itself: the injection point is at the scrubber stack exit, open to the atmosphere, so any HCl produced disperses immediately into the ambient air rather than concentrating against ship surfaces. The only component in contact with the brine solution is the nozzle body. Nickel-chromium-molybdenum alloys such as Inconel 625 resist attack by mineral acids including hydrochloric, nitric, phosphoric, and sulfuric acids in both oxidizing and reducing conditions.[11] Inconel 625 is already used in waste incineration exhaust systems where HCl concentrations are orders of magnitude higher than anything our system would produce. The trace quantities of HCl released in a rapidly diluting open-air exhaust plume pose no credible corrosion threat to either the nozzle or the ship's structure.

3.5 Particle Size, the Plume, and the Brute-Force Approach

The entire MCB field has focused on finding an elegant, energy-efficient solution to the spray problem. This has spiraled in complexity. Salter's design called for 28 billion micro-fabricated silicon nozzles.[20] The University of Washington tested effervescent and supercritical spray nozzles on a decommissioned aircraft carrier before being shut down in Alameda, California in 2024. Cambridge's Centre for Climate Repair is exploring superheated converging-diverging nozzles with shock-wave atomisation. Every approach aims to produce sub-micrometer droplets that dry to the theoretically optimal 30–200 nm salt crystals.[32]

Nephele takes the opposite approach. We propose to do this the "dumb", brute-force way.

The nozzle produces coarse droplets. Simple pressure-atomizing nozzles at 5–20 bar routinely produce droplets of 10–100 µm. This is off-the-shelf industrial fogging hardware, not a research instrument.

The plume does not further atomise droplets. Secondary breakup occurs when the aerodynamic Weber number exceeds $We^{crit} \approx 12$.[34] $We = \rho^{gas} \times v^{rel2} \times d^{droplet} / \sigma$. For post-scrubber exhaust ($\rho^{gas} \approx 0.9$ kg/m³, $v^{rel} \approx 15–30$ m/s, $\sigma^{water} \approx 0.072$ N/m), a 50 µm droplet gives We ≈ 0.14–0.56—far below breakup. The plume will not atomise our droplets further.

The plume does not dry droplets immediately. Post-scrubber exhaust is saturated with water vapour from the scrubbing process. Droplets do not evaporate at the stack exit. However, as the plume rises and entrains ambient marine air (typically 70–80% RH), it dilutes below saturation and evaporation proceeds progressively during the minutes-long transit to cloud base.

The median dry crystal is significantly larger than optimal, but the distribution tail contains the relevant CCN sizes. Per the mass conservation equation: $d^{dry} = d^{droplet} \times (\rho^{sw} \times w^{s} / \rho^{NaCl})^{1/3} \approx 0.255 \times d^{droplet}$. The median 40 µm droplet dries to ~10 µm—well above the optimal CCN range. However, pressure atomizers generally produce a broad lognormal droplet size distribution with a geometric standard deviation $\sigma^{g} \approx 1.8–2.2$.[35] This distribution contains a long tail toward sub-micrometer droplets. A 1 µm droplet dries to a ~255 nm salt crystal; a 0.5 µm droplet to ~128 nm. These are squarely in the optimal CCN-active range.

Quantifying the tail: how many CCN-active particles does one ship produce? Empirically, approximately 0.01–0.1% of the droplet mass from a pressure atomizer falls below 1 µm diameter.[35] Consider a vessel injecting 100 kg NaCl per hour (well within of our operating range discussed in section 3.7). The mass of a single 200 nm dry NaCl particle is approximately 9×10^{-18} kg. If 0.01% of the injected salt mass (0.01 kg/hr) produces particles in the 100–300 nm optimal range, this yields on the order of 10^{15} CCN-active particles per hour; at 0.1%, the yield rises to 10^{16}. The precise fraction depends on nozzle geometry and operating pressure, so 10^{14}–10^{16} is the honest range.

Is that enough to produce a ship track? Pre-2020, ship tracks formed through a very different mechanism: SO^2 gas emitted from the stack slowly oxidised in the atmosphere to form sulfate aerosol particles, primarily in the accumulation mode around 100 nm. This gas-to-particle conversion was slow (typically 1–10% within the first hours downwind) and only a fraction of the resulting particles reached cloud base at the right size.[21] Watson-Parris et al. (2022) catalogued over one million ship tracks from the pre-2020 era and found that track visibility varied enormously between vessels and conditions—many ships produced detectable tracks with modest aerosol perturbations.[9] Our system delivers salt CCN directly into the rising plume as ready-made particles, not as a precursor gas awaiting atmospheric oxidation. This directness is an advantage: every CCN-active particle we inject is immediately available for cloud droplet activation upon reaching cloud base, rather than depending on unpredictable atmospheric chemistry during transit.

The relevant comparison is not the total SO^2 budget of a pre-2020 vessel, but the number of CCN that actually reached cloud base and caused the observed brightening. Ship track observations show that even moderate increases in cloud droplet number concentration (N_d)—on the order of tens of additional droplets per cubic centimetre above the background—are sufficient to produce a detectable albedo perturbation in marine stratocumulus.[19] At 10^{15} CCN-active particles per hour dispersing into a plume that spreads over tens of kilometres downwind, the resulting local N_d perturbation is well within the range observed in historical ship tracks. If anything, the directness of particle delivery (no gas-phase conversion delay) may make our tracks more consistent than the old sulfate tracks, where formation depended on variable atmospheric conditions.

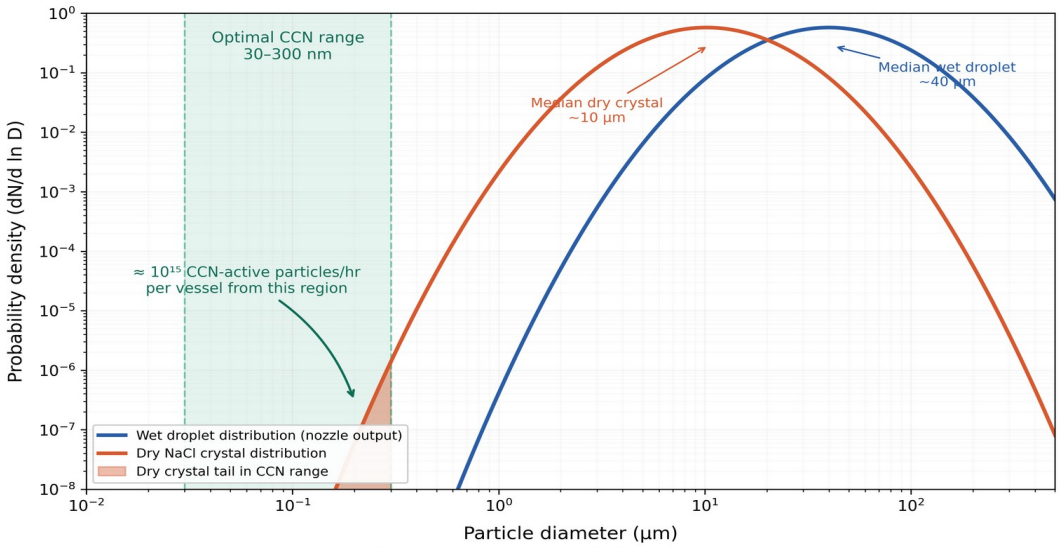

Figure 1. Log-scale view of droplet and crystal size distributions from a pressure-swirl atomizer (Dv50 = 40 µm, σg = 2.0). The log y-axis reveals that the dry crystal distribution (orange) extends well into the CCN-active range (shaded), producing ≈10^{15} particles/hr in the 30–300 nm band per vessel.

Figure 1. Log-scale view of droplet and crystal size distributions from a pressure-swirl atomizer (D_{v50} = 40 µm, σ_g = 2.0). The dry NaCl crystal distribution (orange) is shifted left by the drying factor $d_{dry} \approx 0.255 \times d_{droplet}$. The log y-axis reveals that the crystal distribution tail extends well into the optimal CCN range (30–300 nm, green shading). Although the median crystal (~10 µm) is far above optimal, the sub-micrometer tail yields ≈10^{14}–10^{16} CCN-active particles per hour per vessel—comparable to the effective CCN flux from pre-2020 sulfate ship tracks.

Gravitational settling removes larger particles before they reach cloud base. The plume naturally lofts smaller particles preferentially. A salt crystal's terminal velocity in still air follows the Stokes settling velocity equation:

Particle diameter	Terminal velocity	Time to fall 500 m (MBL depth)
1 µm	~0.007 cm/s	~80 days
5 µm	~0.17 cm/s	~3.4 days
10 µm	~0.66 cm/s	~21 hours
15 µm	~1.5 cm/s	~9 hours

Typical marine boundary layer mixing timescales are on the order of 10–30 minutes, so turbulent eddies can easily loft 1–5 µm particles to cloud base (~500–1000 m). But particles above ~10 µm settle faster than the mean updraft velocity in weakly convective MBL conditions (~0.1–0.5 m/s), meaning they are preferentially removed before reaching the lifting condensation level. At 15 µm, the settling velocity (~1.5 cm/s) is modest in absolute terms but becomes significant over the hours-long residence time needed; these particles also experience enhanced dry deposition and impaction losses near the surface.

The practical implication is that the bulk of the spray mass—which sits in the 5–15 µm range—contributes relatively little to cloud microphysics. The radiatively useful fraction is the sub-5 µm tail, which are the particles most likely to remain suspended long enough to be entrained into cloud updrafts and activate as CCN.

This is wildly energy-inefficient per unit of cloud brightening. Compared to a monodisperse 100 nm spray, our approach wastes the vast majority of injected salt mass on particles too large to be optimal CCN. A dedicated MCB system would be orders of magnitude more efficient. But the energy is already being spent: the ship's engine is running, the scrubber is pumping hundreds of cubic metres of seawater per hour, the exhaust plume is rising to cloud base altitude regardless of whether we inject salt into it. We are piggybacking on systems that will operate this way whether we use them or not. The entire field has been searching for an efficient and elegant solution to marine cloud brightening. We are proposing the inelegant one, because it can be deployed with existing technology on thousands of ships within years, not decades.

3.6 Why Don't Exhaust Scrubbers Already Do This?

A natural question: if the scrubber sprays seawater into the exhaust, why isn't it already brightening clouds? The answer is that the scrubber is specifically designed to prevent exactly this. The scrubber's function is to wash pollutants out of the exhaust gas and into the washwater, which is then discharged to sea. At the scrubber exit, mist eliminators (demister pads) capture entrained water droplets to prevent carryover of washwater into the atmosphere. The scrubber is, in effect, an anti-MCB device: it removes aerosol from the exhaust stream. IMO 2020 compliance depends on this working well.

Nephele's injection point is downstream of the mist eliminator, at the scrubber stack exit. We are adding salt aerosol back into the cleaned exhaust, after the scrubber has done its job of removing sulfur. The scrubber removes the harmful pollutant; we replace the beneficial side effect with a clean substitute. The two systems are complementary, not contradictory.

3.7 How Much Seawater Do We Need?

The SO^2 emission rate from fuel combustion is given by the standard EMEP/EEA emission factor formula:[33]

$$E^{SO2} = 2 \times S \times FC$$

where S is the fuel sulfur mass fraction and FC is fuel consumption (tonnes/day). The factor of 2 is the molecular weight ratio of SO^2 to S (64/32). For a typical large vessel burning FC = 100 tonnes/day of HFO at the pre-IMO 2020 global average sulfur content of S = 0.027 (2.7% by weight):[33]

$$E^{SO2} = 2 \times 0.027 \times 100 = 5.4 \text{ tonnes } SO^2\text{/day}$$

Not all of this SO^2 converted to sulfate aerosol acting as CCN. Atmospheric oxidation is partial and time-dependent, but ship track observations confirm that the resulting sulfate particles were sufficient to produce detectable cloud albedo enhancement.[9]

NaCl has $\kappa \approx 1.28$ versus ammonium sulfate $\kappa \approx 0.61$.[30] However, our particle size distribution is much broader than the fine accumulation-mode sulfate aerosol that formed by gas-to-particle conversion in pre-2020 ship tracks. Giant CCN are less mass-efficient. We conservatively estimate 2–6 tonnes NaCl/day to produce a comparable effective CCN flux, accounting for this size distribution penalty.

At seawater's natural salinity of ~35 g/L, delivering 2–6 tonnes NaCl requires injecting approximately 57–171 m^3 of seawater per day, or roughly 2.4–7.1 m^3/hour. For context, open-loop scrubber systems pump seawater at approximately 45 m^3 per MWh of engine power[33]—a 20 MW vessel circulates over 900 m^3/hr through its scrubber. The additional seawater for salt injection is less than 1% of existing scrubber throughput. Seawater is already filtered for the scrubber and requires no additional treatment.

The control architecture is straightforward: nozzle orifice diameter sets the baseline droplet size distribution; a PID loop modulates brine flow rate, controlling total salt mass delivered per unit time; and an onboard optical particle counter (OPC) downstream measures the actual particle distribution, closing the feedback loop. Satellite data will be used as a longer-timescale feedback input. The nozzle is a simple, replaceable industrial component. The primary tuning variable in operation is how much brine we inject.

4. Verification, Reporting, and Accountability

4.1 Radiative Forcing Verification

The credibility of this proposal rests entirely on transparent, auditable, third-party-verifiable measurement of the radiative forcing produced. We propose the following measurement stack, relying exclusively on established, respected satellite instruments and peer-reviewed methodologies:

Primary: CERES/MODIS Satellite Observations. NASA's Clouds and the Earth's Radiant Energy System (CERES) instruments, flying on Terra, Aqua, Suomi NPP, and NOAA-20 satellites, provide broadband TOA radiative flux measurements validated against global surface observation networks.[13] CERES Energy Balanced and Filled (EBAF) products deliver 1°-resolution monthly mean radiative fluxes constrained to ocean heat storage—the gold standard for Earth radiation budget studies.[14] Coincident MODIS/VIIRS cloud retrievals provide pixel-level cloud droplet number concentration (Nd), effective radius, optical depth, and liquid water path.[15]

Secondary: Ship Track Detection and Attribution. Machine-learning ship track detection algorithms, as demonstrated by Watson-Parris et al. (2022) in cataloguing over one million ship tracks,[9] enable automated identification and attribution of cloud perturbations to specific vessels. AIS (Automatic Identification System) vessel tracking data cross-referenced with satellite overpass times provides a direct chain of attribution from vessel to cloud perturbation to radiative forcing change.

Tertiary: Counterfactual Statistical Methods. Diamond (2023) demonstrated a robust statistical framework for estimating what cloud properties within a shipping corridor would have been absent shipping perturbation.[10] This "what-if" approach provides a rigorous baseline against which to measure the radiative impact of salt aerosol injection.

4.2 Measurement Protocol

1. Each participating vessel logs injection parameters continuously: GPS position, injection rate, brine concentration, ambient meteorological conditions (wind, humidity, boundary layer height from reanalysis).
2. Satellite overpasses (CERES/MODIS, minimum twice daily from Aqua and Terra) are cross-referenced with vessel AIS tracks and injection logs to identify and attribute cloud perturbations.
3. Cloud property changes (ΔNd, $\Delta\tau$, $\Delta albedo$) are measured relative to adjacent unperturbed cloud using both ship-track detection and corridor-scale counterfactual methods.
4. Top-of-atmosphere shortwave flux change is computed per CERES EBAF methodology, yielding radiative forcing in W/m^2 attributable to each vessel.
5. All data (vessel logs, satellite retrievals, attribution analysis) are archived in an open, auditable registry accessible to regulators, independent researchers, and the IMO.

4.3 The Inherent Off Switch

A defining advantage of this approach is reversibility. Sub-micrometer sea salt aerosol in the marine boundary layer has an atmospheric residence time ranging from hours to a few days, depending on particle size, with removal dominated by wet scavenging and gravitational settling.[31] If injection is stopped, the cloud brightening effect ceases within 24–48 hours. There is no persistent atmospheric modification, no long-lived stratospheric loading, no multi-year commitment. This is not stratospheric aerosol injection—it is a short-lived, tropospheric, immediately reversible intervention with real-time satellite verification.

5. What This Is Not

It is important to be explicit about scope. This proposal is:

- Not geoengineering at planetary scale. We are restoring a pre-existing radiative forcing that shipping itself created for decades and then abruptly removed in 2020. We are not adding new forcing; we are replacing what was lost.
- Not new science. The Twomey effect has been studied since 1977. Ship tracks have been observed since Conover (1966). Salt aerosol CCN activation is textbook cloud physics. Corrosion-resistant nickel alloys are mature, commercially available materials. Every component of this proposal is off-the-shelf.
- Not permanent. The atmospheric lifetime of salt aerosol is hours to days. Cessation of injection immediately halts the effect. This is the strongest possible off switch.
- Not sulfur. We replace a harmful pollutant (SO_2/sulfate) with a naturally occurring, ecologically benign substance (NaCl) that has approximately twice the hygroscopicity of sulfate aerosol, meaning it activates as a CCN more readily at equivalent particle size [30]. The health and environmental benefits of IMO 2020 are fully preserved.

6. Advantages Over Dedicated MCB Proposals

Dimension	Dedicated MCB Fleet	Ship-Integrated (Ours)
Vessel infrastructure	Purpose-built spray ships	Existing commercial fleet
Spray system	Precision sub-µm nozzles	Simple pressure atomizers
Vertical transport	Must engineer lofting	Exhaust plume buoyancy
Drying mechanism	Ambient evaporation	Exhaust heat
Capex per unit	$2–5M per spray vessel	~$250K per retrofit
Coverage	Dedicated routes	Global shipping lanes
Off switch	Stop spraying	Stop spraying (same)
Regulatory pathway	Novel; no framework	May fit IMO framework

7. The ZNZ Definition: Where the Door Is Open

The viability of this proposal depends on one regulatory determination: that salt-aerosol cloud brightening qualifies under the IMO Net-Zero Framework's system of rewards and fund disbursement. A close reading of the MEPC 83 text reveals that the door is not just ajar—it was designed to be opened.

7.1 What the Regulation Actually Says

Regulation 31.19 of the draft revised MARPOL Annex VI, as approved at MEPC 83, defines ZNZs as: *"Zero or near-zero GHG emission technologies, fuels and/or energy sources (ZNZs) means the type of ZNZs that qualifies for the annual fuel reward in accordance with regulation 39."*[24]

Three features of this definition are critical:

6. **The tripartite category: "technologies, fuels and/or energy sources."** The ZNZ definition is not limited to fuels. It explicitly includes "technologies" and "energy sources" as standalone qualifying categories. Wind-assisted propulsion, onshore power supply, and onboard carbon capture are already under discussion as potential ZNZ technologies.[25] Salt-aerosol injection is a technology—it is hardware installed on a ship that produces a measurable climate benefit.
7. **The GFI threshold is currently the only criterion.** As multiple analyses have noted, the ZNZ definition at present relies solely on a GHG fuel intensity threshold (≤19.0 gCO^2eq/MJ until 2034; ≤14.0 from 2035). There has been no further differentiation between solutions, whether relating to sustainability criteria or strategic goals.[26] The definition is, by design, "pathway agnostic."[27]
8. **The MEPC retains explicit scope to expand.** The regulation includes a clause providing scope for MEPC to agree on other ZNZs,[25] and the ZNZ reward guidelines are to be defined no later than 1 March 2027 and reviewed every five years thereafter.[24] The details of ZNZ energy sources and technologies, as well as the corresponding compensation amounts, will be determined through future guidelines.[28]

7.2 The Guidelines Gap: Our Window

The MEPC 83 outcome deliberately left the ZNZ guidelines underspecified. As the UMAS/Getting to Zero Coalition analysis stated: "So far, there has been no further differentiation between solutions." Further definition of ZNZs will be included in forthcoming guidelines expected for adoption before 2028.[26] Ten new supporting guidelines and four amended guidelines were identified at MEPC 83 as requiring development.[28] The ICS has submitted draft

guidelines to ISWG-GHG 20 proposing that the ZNZ reward be calculated as a fixed rate per tonne of CO_2eq avoided, using 93.3 gCO_2eq/MJ as the reference baseline.[29]

This creates a specific and time-limited window for intervention. The guidelines that will define which technologies qualify as ZNZ, how rewards are calculated, and how the Net-Zero Fund disburses revenue are being drafted now through ISWG-GHG intersessionals, with most targeted for approval at MEPC 85 (late 2026) alongside the resumed NZF adoption session. There is no final, closed list of qualifying technologies.

7.3 The Argument for Inclusion

Our case for including ship-integrated salt-aerosol injection as a ZNZ technology rests on five points:

- **The framework measures climate impact, not fuel chemistry.** The GFI metric measures gCO_2eq per MJ of energy used. Ship-integrated MCB does not change the fuel; it directly offsets the radiative forcing attributable to the ship's emissions. The net climate effect is equivalent to—or greater than—a fuel switch, and it is independently verifiable.
- **Precedent exists for non-fuel technologies.** Wind-assisted propulsion, solar power, and onshore power supply are already being discussed as ZNZ-eligible technologies that reduce a ship's GFI without changing the fuel burned. Onboard carbon capture and storage (OCCS) received an approved work plan at MEPC 83 for regulatory framework development.[28] These technologies reduce net emissions without being fuels. Salt-aerosol injection reduces net radiative forcing without being a fuel. The logic is identical.
- **The framework's own origin story demands it.** The IMO 2020 sulfur regulation—implemented under this same MARPOL Annex VI—removed the inadvertent cloud-brightening effect. The Net-Zero Framework exists in part because of the urgency created by shipping's warming contribution. Allowing ships to restore the lost cooling via clean salt aerosol is not a stretch of the framework; it is a direct remedy for a problem the framework was created to address.
- **The Fund's mandate is broad.** Regulation 41 of the draft Annex VI specifies that the IMO Net-Zero Fund shall disburse revenue for rewarding ZNZ uptake, promoting just transition, and facilitating environmental and climate protection, adaptation and resilience building.[17] Verified radiative forcing from salt aerosol is climate protection in its most literal form.
- **It restores what the IMO itself removed.** This is not geoengineering; it is regulatory remediation. The IMO created the cooling gap through IMO 2020. Allowing the Net-Zero Fund to pay for restoring that cooling closes a circle that the IMO's own regulations opened.

7.4 What We Need from MEPC

Specifically, we need the MEPC to do one of the following in the ZNZ guidelines currently under development:

9. **Option A (preferred):** Explicitly include “sea-salt aerosol ship track remediation” as a named eligible ZNZ technology category—defined as the injection of sea salt aerosol from registered commercial vessels into the marine boundary layer via existing exhaust infrastructure, for the purpose of restoring cloud-albedo radiative forcing lost following IMO 2020 sulfur regulations. Radiative forcing to be measured in W/m^2 and converted to CO^2eq-avoided using standard climate equivalence metrics. This narrow definition ensures that only ship-integrated, salt-based, satellite-verified interventions qualify—not unregulated open-ocean spraying or novel atmospheric modification by non-shipping actors.
10. **Option B:** Define ZNZ technology eligibility broadly enough to encompass any ship-installed technology that produces a verified net negative radiative forcing, without restricting the category to fuel-switching or energy-efficiency measures.
11. **Option C (fallback):** Create a separate Fund disbursement category under Regulation 41’s climate protection mandate for verified radiative-forcing interventions, distinct from the GFI-based ZNZ reward but funded from the same Net-Zero Fund revenue pool.

All three options are consistent with the approved regulatory text. None requires amending the draft MARPOL Annex VI—only specifying the guidelines that MEPC is already mandated to develop.

8. Regulatory Pathway and Timeline

The regulatory calendar as of March 2026 creates natural insertion points for our proposal:

Date	Action
27 Apr–1 May 2026 (MEPC 84)	Upcoming. Nephele attends for corridor conversations, delegation engagement, and side-event presence. MEPC 84 will address non-NZF MARPOL amendments and set the date for the resumed extraordinary session.
Oct/Nov 2026 (MEPC/ES.2 resumed)	Target for formal NZF adoption. Draft legal text dates to be updated. ZNZ guidelines targeted for approval at MEPC 85 (also late 2026).
Early 2027	Original deadline for MEPC to define ZNZ reward level and methodology (will shift if adoption is in late 2026).
1 Jan 2028 (earliest)	First reporting year under NZF. First GFI compliance obligations apply.
Mid-2029	First reward payments disbursed from Net-Zero Fund.
2033	Five-year review of NZF including ZNZ definitions.

We propose a phased approach aligned to this calendar:

12. Pilot phase (Q4 2026–Q4 2027, 10–20 vessels): Demonstrate hardware reliability on scrubber-equipped vessels; validate satellite measurement protocol; publish results in peer-reviewed literature. Run concurrently with ISWG-GHG guideline development.
13. MEPC engagement: Attend MEPC 84 (April 2026) for delegation engagement and side-event presence. Target formal paper submissions to ISWG-GHG intersessionals (submission windows open mid-2026) and MEPC 85/resumed ES.2 (late 2026). Propose ZNZ guideline language under Option A, B, or C above. Engage flag states, ICS, and Pacific Island nations as sponsors.
14. Scale-up (2028+): Deploy across scrubber-equipped fleet once ZNZ recognition is secured, targeting corridors with high marine stratocumulus prevalence (Northeast Pacific, Southeast Atlantic, Southern Ocean).

8.5 MEPC Oversight as Strategic Position

Nephele is not seeking to operate outside the IMO’s regulatory framework. We are seeking to operate inside it—with full MEPC oversight, under rules the MEPC itself defines. We will work with the Committee to develop the regulatory standards for ship-based salt-aerosol radiative forcing, including verification methodology, operational limits, corridor restrictions, and cessation protocols. We are proposing to give the IMO direct governance authority over this activity.

This is a deliberate strategic choice. The hardware Nephele proposes—simple nozzles, a flow valve, a particle counter—is not proprietary in any meaningful sense. The satellite data used for verification (CERES, MODIS, VIIRS) is publicly available. There is no deep technology moat in the conventional sense. What Nephele seeks to build is a regulatory moat: the institutional relationship with the MEPC, the approved verification methodology, the track record of auditable operations, and the trusted-operator status that comes from being the entity that helped write the rules and then operated transparently under them. A competitor could replicate the hardware. They cannot easily replicate a multi-year relationship with the IMO built on demonstrated compliance, published data, and institutional trust.

By inviting MEPC oversight, Nephele also addresses the governance concerns that have stalled other MCB efforts. The Alameda shutdown, the broad civil society opposition to unregulated geoengineering experiments—these are fundamentally objections to unaccountable actors operating without institutional oversight. Our proposal is the opposite: a regulated activity, on registered vessels, under international treaty authority, with satellite-verified accountability and an institutional off switch controlled by the MEPC itself.

9. Revenue Model

9.1 How the Money Flows

The IMO Net-Zero Framework was approved in principle at MEPC 83 in April 2025 but its formal adoption was adjourned at the extraordinary MEPC session in October 2025, following US opposition and a vote to defer by 57–49. The resumed extraordinary session is expected in late 2026, with entry into force no earlier than March 2028.[16] Once operative, the framework will introduce mandatory GHG fuel intensity (GFI) targets for ships over 5,000 GT and a pricing mechanism for non-compliance. Ships exceeding GFI targets must purchase Remedial Units; ships outperforming them generate tradeable Surplus Units. Revenue flows into the IMO Net-Zero Fund, which is mandated to reward zero and near-zero emission technologies, fund the energy transition, and support vulnerable states.[17]

Our proposal fits this framework as follows:

15. The IMO recognises verified ship-track radiative forcing as a qualifying climate intervention eligible for support from the Net-Zero Fund.
16. Participating shippers install salt injection modules on their scrubber-equipped vessels and operate them along designated corridors.
17. Radiative forcing is verified via the satellite measurement protocol described in Section 4.
18. Nephele invoices the IMO Net-Zero Fund for verified radiative forcing delivered, priced per $W/m^2 \cdot km^2 \cdot day$ (or equivalent unit).
19. Nephele pays participating shippers a per-voyage fee to operate the injection equipment, offsetting a portion of their GHG compliance costs under the same IMO framework.

9.2 Circularity

This creates a self-sustaining circular economy:

- Shippers who are already paying GHG compliance costs under the Net-Zero Framework receive payment for operating injection equipment, reducing their net compliance burden.
- The IMO Net-Zero Fund deploys capital toward a verified, measurable, immediately reversible climate intervention—exactly the type of technology transition it was created to support.
- Nephele sustains operations from the margin between Fund payments received and shipper fees paid.
- The climate benefits are real, satellite-verified, and independently auditable—not offset certificates, not projections, but measured changes in Earth’s energy balance.

At scale, with the IMO Net-Zero Fund expected to generate $10–12 billion annually in its first three years,[18] even a small allocation toward verified radiative forcing interventions would fund deployment across hundreds of vessels.

9.3 Unit Economics

Cost Item	Estimate
Hardware, installation, and commissioning (per vessel)	~$250,000 (estimated)
Consumables (nozzle replacement, per year)	$1,000–$3,000
Pump energy (per day at sea)	<$10 (< 2 kW)
Verification infrastructure (shared)	Amortised across fleet

Estimated per-vessel cost is approximately $250,000 including hardware, installation, and commissioning. This figure is preliminary and will be refined during prototype development and sea trials. For context, a single scrubber installation costs $2–8 million. The injection module represents a fraction of existing scrubber investment, and annualised operating costs (consumables, energy) are under $5,000 per vessel.

9.4 Funding Pathway: From Regulatory Intent to Deployment Capital

NZF disbursements will not begin until the Framework enters into force (March 2028 at earliest). The engineering development, prototyping, and pilot deployment required before that date must be funded independently. With public IMO support and a clear projected revenue stream from NZF payments, securing venture capital for this pre-revenue phase becomes straightforward. The VC proposition is: simple hardware, a defined regulatory pathway, an institutional partner in the MEPC, projected per-vessel revenue from a $10–15 billion/year fund, and a regulatory moat that compounds over time. This is a fundable thesis.

Additionally, with public IMO backing, Nephele can accept payments from other climate funds and public grants to subsidise hardware installation for shippers. The Green Climate Fund, bilateral climate finance mechanisms, and national maritime decarbonisation programmes all represent potential funding sources that become accessible once the IMO has formally recognised the activity as a legitimate climate intervention. These alternative funding streams can accelerate deployment before NZF payments begin and reduce the capital burden on early-adopting shipowners.

10. Risks and Mitigations

Risk	Severity	Mitigation
Nozzle fouling / blockage	Low	Replaceable nozzle components; filtered seawater intake; redundant nozzle array
Insufficient plume lofting	Low	Ship exhaust buoyancy is well-characterised; decades of ship track observations confirm boundary layer penetration
Cloud response varies by regime	Medium	Target deployment in stratocumulus-dominant corridors; avoid regions where cloud dissipation may dominate (open-cell convection)
IMO does not recognise intervention	High	Primary regulatory risk; mitigated by strong scientific evidence base and alignment with Net-Zero Fund mandate
Public opposition to “geoengineering”	Medium	Frame accurately: restoration of pre-existing effect with benign material; emphasise reversibility and satellite transparency
Corrosion of scrubber components	Low	Injection is downstream of scrubber; salt spray contacts only corrosion-resistant nozzle alloy and open atmosphere

11. Conclusion

The removal of sulfate aerosol from shipping emissions has created a measurable warming gap. The physics of cloud brightening via salt aerosol is established. The delivery mechanism—ship exhaust plumes—is already operating across every ocean. The hardware required is simple, cheap, and uses proven materials. The verification methodology relies on the world’s most trusted satellite radiometric instruments. The off switch is immediate and total.

The only barrier is regulatory recognition. The IMO Net-Zero Framework, with its dedicated fund and mandate to support climate-positive technologies, provides the institutional vehicle. What we need is the political will to deploy a solution that is already, in every technical sense, ready.

We are not proposing to do anything new. We are proposing to do what ships already did for decades—brighten clouds along their wakes—but deliberately, cleanly, verifiably, and with a substance that belongs in the ocean.

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